

SAP ID: 60019220125

Branch: CSE(ICB)

EXPERIMENT 2

Aim: Installation of Androguard and analysis on apk files.

Screenshots:

Step 1:

```
File Actions Edit View Help

Summary:
Upgrading: 44, Installing: 17, Removing: 2, Not Upgrading: 2083
Download size: 19.8 MB / 52.9 MB
Space needed: 79.6 MB / 61.7 GB available

Get:1 http://kati.kali.org/kali kali-rolling/main amd64 libpython3.12-dev amd64 3.12.8-5deb1 [5,129 kB]
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```

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Downloading androguard-4.1.2-py3-none-any.whl (950 kB)
950.2/950.2 kB 6.4 MB/s eta 0:00:00
Downloading apkinspector-1.3.2-py3-none-any.whl (29 kB)
Downloading dataset-1.6.2-py2.py3-none-any.whl (18 kB)
Downloading frida-16.6.6-cp37-ab13-manylinux2_5_x86_64.whl (29.8 MB)
29.8/29.8 MB 6.2 MB/s eta 0:00:00
Downloading loguru-0.7.3-py3-none-any.whl (61 kB)
61.6/61.6 kB 335.2 kB/s eta 0:00:00
Downloading banal-1.0.6-py2.py3-none-any.whl (6.1 kB)
Building wheels for collected packages: mtf8
  Building wheel for mtf8 (setup.py) ... done
  Created wheel for mtf8: filename=mtf8-1.0.6-cp311-cp311-linux_x86_64.whl size=19248 sha256=a3ab7bb7fc9b253f00d6b116f736ad09f0bc8a410b70e1511cff713683ba2c
  Stored in directory: /home/kali/.cache/pip/wheels/44/bc/d6/647be2901ee64bf5bbc7636ea53f146947a204f61d552771d3
Successfully built mtf8
Installing collected packages: mtf8, banal, loguru, frida, apkInspector, dataset, androguard
WARNING: The script apkInspector is installed in '/home/kali/.local/bin' which is not on PATH.
Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
WARNING: The script androguard is installed in '/home/kali/.local/bin' which is not on PATH.
Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
Successfully installed androguard-4.1.2 apkInspector-1.3.2 banal-1.0.6 dataset-1.6.2 frida-16.6.6 loguru-0.7.3 mtf8-1.0.6

```

Step 2: analyze



Name: Gargi Sukhatankar

SAP ID: 60019220125

Roll No: B027

Branch: CSE(ICB)

```
root@kali: /home/kali
File Actions Edit View Help
(kali@kali)~[*]
$ sudo su
[sudo] password for kali:
(root@kali)~[/home/kali]
(root@kali)~[/home/kali]
# androguard analyze /home/kali/Downloads/dr-driving-mod-1.72-drdrivngapk.com.apk
Please be patient, this might take a while.
Found the provided file is of type 'APK'
[INFO ] androguard.apk: Starting analysis on AndroidManifest.xml
[INFO ] androguard.apk: APK file was successfully validated!
[WARNING] androguard.core.api_specific_resources: Requested API level 34 is larger than maximum we have, returning API level 28 instead.
[INFO ] androguard.analysis: Adding DEX file version 35
[INFO ] androguard.analysis: Reading bytecode took : 0min 07s
[INFO ] androguard.analysis: Adding DEX file version 35
[INFO ] androguard.analysis: Reading bytecode took : 0min 01s
[INFO ] androguard.analysis: End of creating cross references (XREF) run time: 0min 09s
Added file to session: SHA256::22bfed7fbb20b7d96a5c674147feae66a316585d636ab3e4c73fc63c485bb1ed
Loaded APK file ...
>>> a
<androguard.core.bytecodes.apk.APK object at 0x7f5282150090>
>>> d
[<androguard.core.bytecodes.dvm.DalvikVMFormat object at 0x7f528219c790>, <androguard.core.bytecodes.dvm.DalvikVMFormat object at 0x7f5254a2de50>]
>>> dx
<analysis.Analysis VMs: 2, Classes: 17042, Methods: 86253, Strings: 21794>
Androguard version 3.4.0a1 started
In [1]:
```

```
In [1]: a
Out[1]: <androguard.core.bytecodes.apk.APK at 0x7f5282150090>

In [2]: d
Out[2]:
[<androguard.core.bytecodes.dvm.DalvikVMFormat at 0x7f528219c790>,
 <androguard.core.bytecodes.dvm.DalvikVMFormat at 0x7f5254a2de50>]

In [3]: dx
Out[3]: <analysis.Analysis VMs: 2, Classes: 17042, Methods: 86253, Strings: 21794>

In [4]: a.get_permissions()
Out[4]:
['com.google.android.providers.gsf.permission.READ_GSERVICES',
 'android.permission.ACCESS_NETWORK_STATE',
 'com.android.vending.BILLING',
 'android.permission.ACCESS_AD_SERVICES_ATTRIBUTION',
 'android.permission.ACCESS_AD_SERVICES_TOPICS',
 'android.permission.FOREGROUND_SERVICE',
 'com.ansangha.drdriving.DYNAMIC_RECEIVER_NOT_EXPORTED_PERMISSION',
 'android.permission.INTERNET',
 'android.permission.ACCESS_AD_SERVICES_AD_ID',
 'android.permission.WAKE_LOCK',
 'com.google.android.finsky.permission.BIND_GET_INSTALL_REFERRER_SERVICE',
 'com.google.android.gms.permission.AD_ID']

In [5]: a.get_activities()
Out[5]:
['com.ansangha.drdriving.DrDrivingActivity',
 'com.google.android.gms.ads.AdActivity',
 'com.android.billingclient.api.ProxyBillingActivity',
 'com.android.billingclient.api.ProxyBillingActivityV2',
 'com.google.android.gms.auth.api.signin.internal.SignInHubActivity',
 'com.google.android.gms.games.internal.v2.resolution.GamesResolutionActivity',
 'com.google.android.gms.games.internal.v2.appshortcuts.PlayGamesAppShortcutsActivity',
 'com.google.android.gms.common.api.GoogleApiActivity',
 'com.google.android.gms.ads.OutOfContextTestingActivity',
 'com.google.android.gms.ads.NotificationHandlerActivity']

In [6]: a.get_package()
Out[6]: 'com.ansangha.drdriving'

In [7]: a.get_app_name()
Out[7]: 'Dr. Driving'

In [8]: a.get_app_icon()
Out[8]: 'res/drawable-xxxhdpi/ic_launcher.png'

In [9]: a.get_androidversion_code()
Out[9]: '72'
```

Step 3: Decompile



**SHRI VILEPARLE KELAVANI MANDAL'S
DWARKADAS J. SANGHVI COLLEGE OF ENGINEERING**
(Autonomous College Affiliated to the University of Mumbai)
NAAC ACCREDITED with "A" GRADE (CGPA : 3.18)



Name: Gargi Sukhatankar

SAP ID: 60019220125

Roll No: B027

Branch: CSE(ICB)

```
root@kali: ~/home/kali
$ androguard decompile -o decompiled_code /home/kali/Downloads/dr-driving-mod-1.72-drdrivingapk.com.apk
[INFO ] androguard.apk: Starting analysis on AndroidManifest.xml
[INFO ] androguard.apk: APK file was successfully validated!
[WARNING] androguard.core.api_specific_resources: Requested API level 34 is larger than maximum we have, returning API level 28 instead.
[INFO ] androguard.analysis: Adding DEX file version 35
[INFO ] androguard.analysis: Reading bytecode took : 0min 07s
[INFO ] androguard.analysis: Adding DEX file version 35
[INFO ] androguard.analysis: Reading bytecode took : 0min 02s
[INFO ] androguard.analysis: End of creating cross references (XREF) run time: 0min 10s
Dump information /home/kali/Downloads/dr-driving-mod-1.72-drdrivingapk.com.apk in decompiled_code
Create directory decompiled_code
Decompilation ... End
Dump La/a; <init> (V ... source codes ... bytecodes ...
Dump La/a; a (Ljava/lang/StackTraceElement; ... bytecodes ...
Dump La/b; <clinit> (V ... source codes ... bytecodes ...
Dump La/b; b (Ljava/lang/Throwable; Ljava/lang/String;)Ljava/lang/StackTraceElement; ... bytecodes ...
Dump La0/a$a; <init> (La0/a; Lz/e;)V ... source codes ... bytecodes ...
Dump La0/a$a; newCursor (Landroid/database/sqlite/SQLiteDatabase; Landroid/database/sqlite/SQLiteCursorDriver; Ljava/lang/String; Landroid/database/sqlite/SQLiteQuery;)Landroid/database/Cursor; ... bytecodes ...
Dump La0/a$a; <init> (La0/a; Lz/e;)V ... source codes ... bytecodes ...
Dump Lz/b; AB (Lz/e;)Landroid/database/Cursor; ... source codes ... bytecodes ...
Dump Lz/b; F (Ljava/lang/String;)Lz/f; ... bytecodes ...
Dump Lz/b; N (Lz/e; Landroid/os/CancellationSignal;)Landroid/database/Cursor; ... bytecodes ...
Dump Lz/b; Y (Ljava/lang/String; Ljava/lang/Object;)V ... bytecodes ...
Dump Lz/b; a (V ... bytecodes ...
Dump Lz/b; d (V ... bytecodes ...
Dump Lz/b; e0 (Ljava/lang/String;)Landroid/database/Cursor; ... bytecodes ...
Dump Lz/b; getPath ()Ljava/lang/String; ... bytecodes ...
Dump Lz/b; h (V ... bytecodes ...
Dump Lz/b; isOpen ()Z ... bytecodes ...
Dump Lz/b; s0 ()Z ... bytecodes ...
Dump Lz/b; t ()Ljava/util/List; ... bytecodes ...
Dump Lz/b; w (Ljava/lang/String;)V ... bytecodes ...
Dump La0/a; <clinit> (V ... source codes ... bytecodes ...
Dump La0/a; <init> (Landroid/database/sqlite/SQLiteDatabase;)V ... bytecodes ...
```